

Test 18: Chiral symmetry AIII at $\nu=1/2$

Verdict: PASS | Generated: 2026-09-02 15:12:06 | Suite: AB-Cloud v19 (Julia 1.12.0)

What this test verifies

EXACT: $\|SHS+H\|/\|H\| \lesssim 1e-16$ on the pure lattice ($W=0$); $W>0$ is the honest breaker.

METHOD: EXACT: $\|SHS+H\|/\|H\| \lesssim 1e-16$ on the pure lattice ($W=0$); $W>0$ is the honest breaker.

Test 18 HARDCORE: exact 0.00e+00, ν -row 5.77e-03 (ν_{rms} 0.0058, in window), vortex 0.00e+00 (preserved), $S^2=I$ 0.00e+00 (96x96 torus).

Result [HARDCORE pass 2]: 4 sub-checks, 0 failed $\Rightarrow$ PASS

- 'plots/plot_01' (.png 600dpi / .svg / .pdf)

Plots

- 'plots/animation.gif' $\Rightarrow$ animated reveal of every plot of this test

Reproduce with: `julia ab_cloud_v19.jl --test 18 --no-two-pass`

Independent verification guide

Recompute over logs in logs/computation_log.txt (timestamps + deltas); raw console output in logs/stdout_capture.txt.

[15:11:28.198] [+5003.481s] build_ab_cloud_hamiltonian: 96x96 (N=9216), $\nu=0.5000$, t=1.00, W=0.00, torus, 0 vortices, model=:monumental

[15:11:40.131] [+5015.413s] build_ab_cloud_hamiltonian: 96x96 (N=9216), $\nu=0.5000$, t=1.00, W=1.00, torus, 0 vortices, model=:monumental

FULL COMPUTATION LOG

[15:11:52.929] [+5028.211s] build_ab_cloud_hamiltonian: 96x96 (N=9216), $\nu=0.5000$, t=1.00, W=0.00, torus, 2 vortices, model=:monumental

??

TEST 18 (pass 2, HARDCORE) - chiral symmetry AIII $\Rightarrow$ deep audit

CONSOLE CAPTURE

Exact row + ν_{rms} quantitative contract + vortex-preservation row +

$S^2 = I$ operator sanity (launch-standard lattice)

??

ν RUNTIME: 3 builds + 3 S+H+S defect evals at 96x96 (Diagonal S, $O(N^?)$) $\Rightarrow$? 1?2 min

H1 exact row ($\nu=1/2$, W=0): defect = 0.00e+00 $<$? 1e-10

H2 ν row (W=1.0, Nv=0): defect = 5.77e-03 vs ν_{rms} = 0.0058 (analytic) $\Rightarrow$ [0.5, 2.0]x?

H2v vortex row (Nv=2, W=0): defect = 0.00e+00 $<$ 1e-10 $\Rightarrow$ (bond phases PRESERVE AIII)

H3 operator sanity: $\nu S^2 \nu$ I/?I? = 0.00e+00 $<$? 1e-12

??

exact row $<$ 1e-10 PASS $\Rightarrow$ $\nu SHS+H$?/?H? = 0.00e+00 at $\nu=1/2$, W=0

ν_{rms} contract (W=1.0) PASS $\Rightarrow$ defect 5.77e-03 vs analytic 2???_F/?H?_F = 0.0058 (size-independent)

vortices PRESERVE AIII PASS $\Rightarrow$ defect = 0.00e+00 (Nv=2, q=-1.00, W=0) $\Rightarrow$ bond phases; Byers-Yang for integer q

S? = I PASS ? ?S??I?/?I? = 0.00e+00

Test 18b [HARDCORE pass 2]: 4 sub-checks, 0 failed ? PASS